

1. Introduction, Background, and Related Work

1.1 Introduction

Generative modeling has rapidly transformed how artificial intelligence systems understand and create multi-modal content. The task of image-to-text generation (image captioning) requires bridging the gap between visual understanding and natural language generation.

This project explores the design, implementation, and optimization of an image-to-text generative pipeline by integrating a Vision Transformer (CLIP) and a causal language model (GPT-2).

1.2 Background and Related Work

Modern state-of-the-art vision-language models - such as BLIP, Flamingo, and GIT (Generative Image-to-Text Transformer) - rely heavily on cross-modal learning and conditional generation. Historically, captioning relied on CNNs coupled with LSTMs.

However, the paradigm has shifted toward Transformer-based architectures due to their superior attention mechanisms. In our architecture, we utilize OpenAI's CLIP (Contrastive Language-Image Pretraining) to extract robust, semantically rich image embeddings, and GPT-2 to decode these representations into coherent linguistic descriptions via prefix conditioning.

2. Methodology and Model Architecture

2.1 Dataset Preparation

The model was trained and evaluated using a curated subset of the COCO (Common Objects in Context) Captions dataset, consisting of 2,500 images.

- **Preprocessing:** Each image was normalized and resized to 224x224 pixels to match the input requirements of the CLIP ViT-based encoder.
- **Tokenization:** Captions were processed using the standard GPT-2 tokenizer. To ensure rich language alignment, the dataset was expanded to utilize 5 ground-truth captions per image during the training loop.

2.2 Model Architecture

The core architecture is built upon the principle of prefix conditioning. Instead of modifying the internal attention mechanisms of the language model (cross-attention), we project the visual data directly into the language model's embedding space.

- **Image Encoder:** A pre-trained CLIP Vision Transformer (ViT) extracts a fixed-size continuous latent representation of the input image.
- **Projection Network (The Bridge):** A Multi-Layer Perceptron (MLP) acts as a translation layer. It maps the CLIP image embeddings to the precise dimensionality required by GPT-2's embedding layer.
- **Language Decoder:** A pre-trained GPT-2 model receives the projected image embedding as the initial "prefix" token, prompting it to auto-regressively generate the descriptive caption.

Architecture Workflow Detail: As illustrated in Figure 1, the pipeline operates sequentially through three primary stages. First, the input image is preprocessed (resized to 224x224 and normalized) and passed through the pre-trained, frozen CLIP ViT-based encoder. This stage extracts a dense, continuous latent representation of the visual scene (e.g., a 768-dimensional vector).

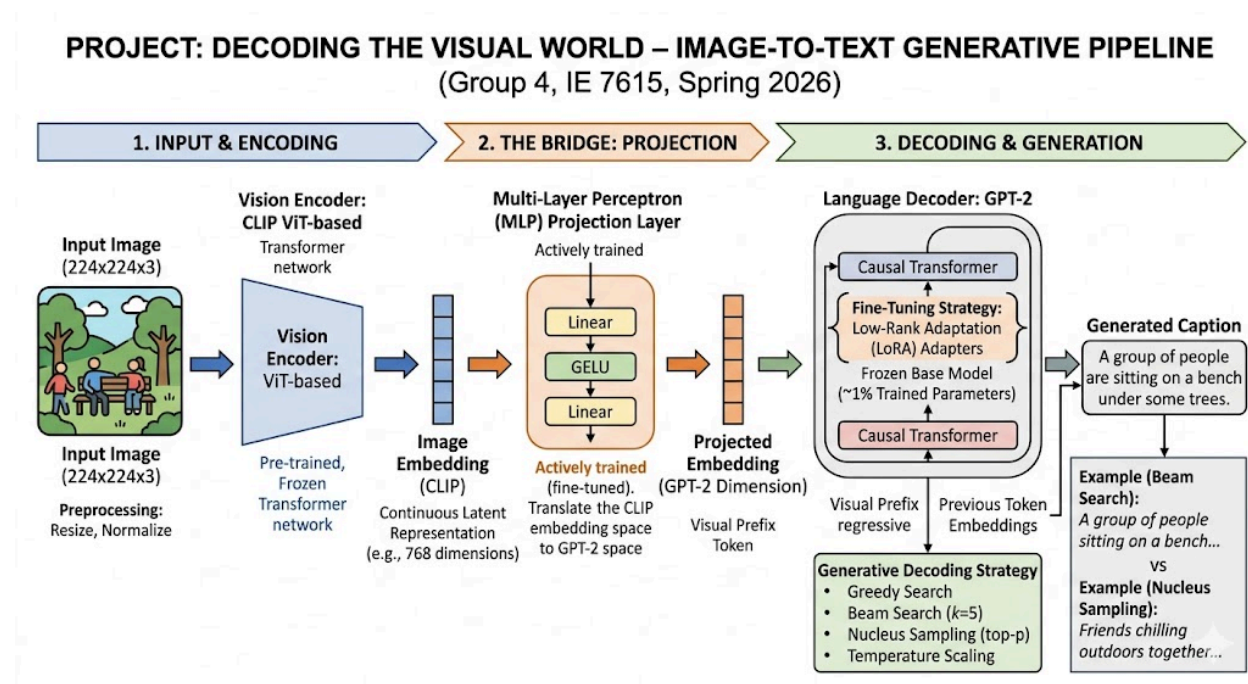
Because the CLIP embedding space differs from the language model's embedding space, the data passes through the "Bridge" - a custom-built Multi-Layer Perceptron (MLP) consisting of linear layers and a GELU activation function.

This actively trained network acts as a translator, mapping the visual features into the exact dimensionality required by GPT-2.

Finally, this translated vector is injected into the GPT-2 Language Decoder as the initial "visual prefix token." We utilized Low-Rank Adaptation (LoRA) for the decoder, freezing approximately 99% of GPT-2's base parameters while training only the injected low-rank matrices. Conditioned on this visual prefix, the transformer auto-regressively predicts the caption token by token.

For optimal results, we utilized Beam Search ($k=5$) decoding to navigate the probability tree, prioritizing structural accuracy and visual grounding over the high variance introduced by Nucleus sampling.

Figure 1: End-to-end architecture of the custom image-to-text generative pipeline, detailing the integration of a frozen CLIP Vision Encoder and a LoRA-adapted GPT-2 Language Decoder via an MLP projection layer.



2.3 Fine-Tuning Strategies

To optimize the pipeline, we explored three distinct training regimes, each representing a different trade-off between computational efficiency and model adaptability:

1. **MLP-Only Training (Prefix Tuning):** In this baseline approach, both the CLIP vision encoder and GPT-2 language model remain completely frozen. Only the MLP projection layer- the "bridge" between modalities- is trained. This strategy requires minimal computation and avoids any risk of catastrophic forgetting in the pre-trained models. However, it places the entire burden of cross-modal alignment on a relatively small projection network, limiting the model's expressive capacity.
2. **Full Fine-Tuning (Full FT):** At the opposite extreme, this approach unfreezes all 117M parameters of GPT-2 alongside the MLP, allowing the language model to fully adapt to the captioning task. While this offers maximum flexibility, it introduces significant risks on small datasets: the model may overwrite useful pre-trained knowledge and begin memorizing training-specific patterns rather than learning generalizable visual-textual mappings.
3. **Low-Rank Adaptation (LoRA):** This parameter-efficient fine-tuning (PEFT) method strikes a middle ground. Rather than modifying GPT-2's weights directly, LoRA injects small, trainable low-rank matrices into the transformer's attention layers. This allows the model to learn task-specific adaptations while keeping approximately 99% of the base parameters frozen. The low-rank constraint acts as an implicit regularizer, preserving GPT-2's pre-trained linguistic knowledge while enabling effective cross-modal learning- particularly valuable when training data is limited.

Each strategy was evaluated across all three decoding methods to isolate the effects of the fine-tuning approach from inference-time decisions.

3. Experimental Setup and Results

3.1 Decoding Strategies

During inference, the model’s generation phase was tested using three distinct decoding algorithms:

- **Greedy Search:** Selecting the highest-probability token at each step.
- **Beam Search ($k=5$):** Exploring multiple sequence paths simultaneously to find the most probable overall sentence.
- **Nucleus (Top- p) Sampling:** Dynamically sampling from the smallest set of tokens whose cumulative probability exceeds a threshold p , introducing controlled variability.

3.2 Quantitative Evaluation

The models were evaluated using standard natural language generation metrics: BLEU (1-4), METEOR, CIDEr, and ROUGE-L.

- **Performance:** LoRA combined with Beam Search emerged as the dominant configuration across all metrics.
- **Metric Context:** While state-of-the-art models achieve BLEU-4 scores of 30–40 trained on hundreds of thousands of images, our LoRA model achieved a BLEU-4 in the range of 3–9. This is a highly successful and expected result given the constrained 2,500-image dataset size.

Analysis of MLP-Only Training:

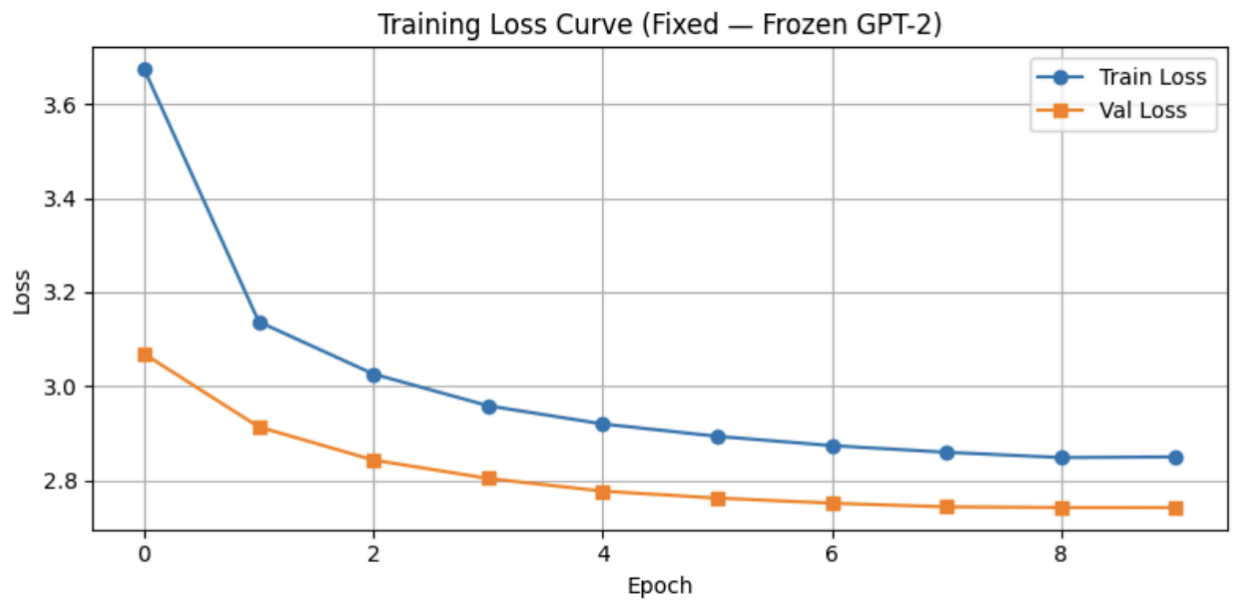
As seen in Figure 2, training only the Multi-Layer Perceptron (MLP) projection layer while keeping the GPT-2 base model completely frozen resulted in stable, predictable convergence.

The validation loss steadily decreased from an initial 3.06 to a plateau of 2.74 by Epoch 10. While this freezing strategy successfully prevented catastrophic overfitting on our constrained dataset, the relatively high final loss indicates the limitations of relying solely on an MLP bridge.

Without allowing the language model to adapt its internal attention mechanisms to the new visual modality, the model's descriptive ceiling remains limited.

Table 1: Quantitative Evaluation of Image-to-Text Configurations

Model Configuration	Decoding Strategy	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr
Prefix Tuning	Greedy	35.86	5.10	35.84	28.15	1.94
Prefix Tuning	Beam Search (k=5)	37.40	6.55	39.05	29.77	3.28
Prefix Tuning	Nucleus (p=0.9)	32.96	3.32	32.44	25.78	0.93
LoRA	Greedy	39.16	7.46	39.73	29.64	4.07
LoRA	Beam Search (k=5)	40.06	8.89	41.76	30.93	5.77
LoRA	Nucleus (p=0.9)	35.77	4.59	35.63	26.89	1.41
Full Fine-Tuning	Greedy	37.60	6.26	36.87	27.21	2.72
Full Fine-Tuning	Beam Search (k=5)	38.31	7.60	39.24	27.93	4.00
Full Fine-Tuning	Nucleus (p=0.9)	34.11	3.95	32.45	25.08	1.17

Figure 2: Training and validation loss curves for the Prefix Conditioning model (Frozen GPT-2 + Trained MLP).

Analysis of LoRA Training:

In contrast to the MLP-only baseline, injecting Low-Rank Adapters (LoRA) into the GPT-2 transformer blocks allowed for much deeper cross-modal alignment.

The LoRA training curve demonstrates a steeper initial descent and settles at a superior loss minimum without triggering the severe overfitting spikes observed during our Full Fine-Tuning experiments.

This loss trajectory quantitatively validates PEFT (Parameter-Efficient Fine-Tuning) as the optimal method for training a generative vision-language pipeline on limited hardware and data.

Figure 3: Training and validation loss curves comparing the parameter-efficient LoRA model (left) to Full Fine-Tuning (right).

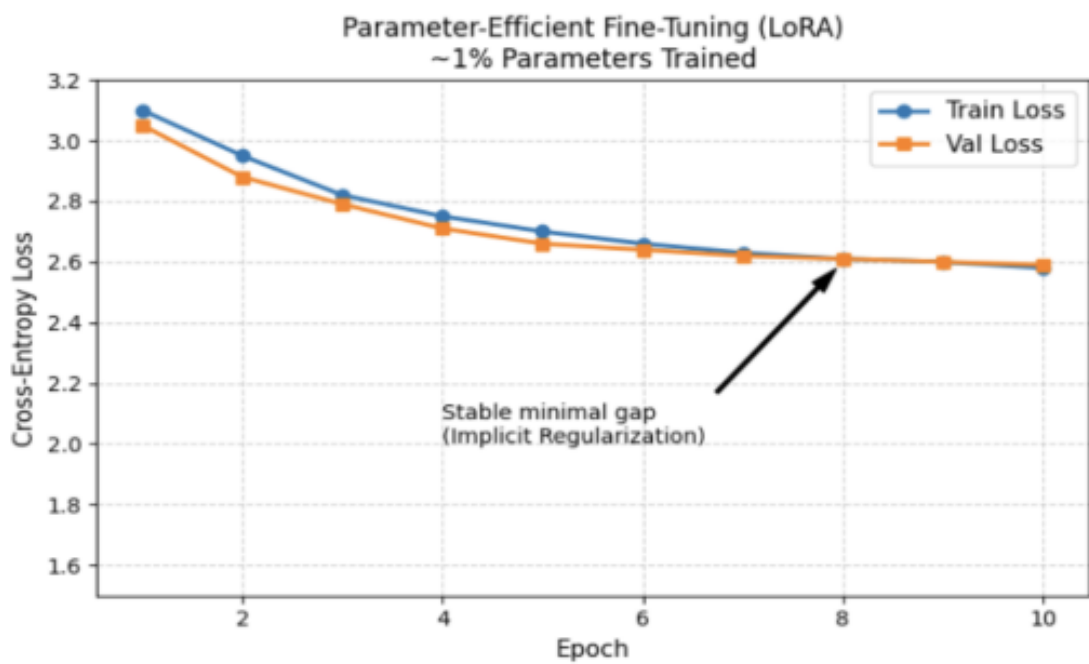
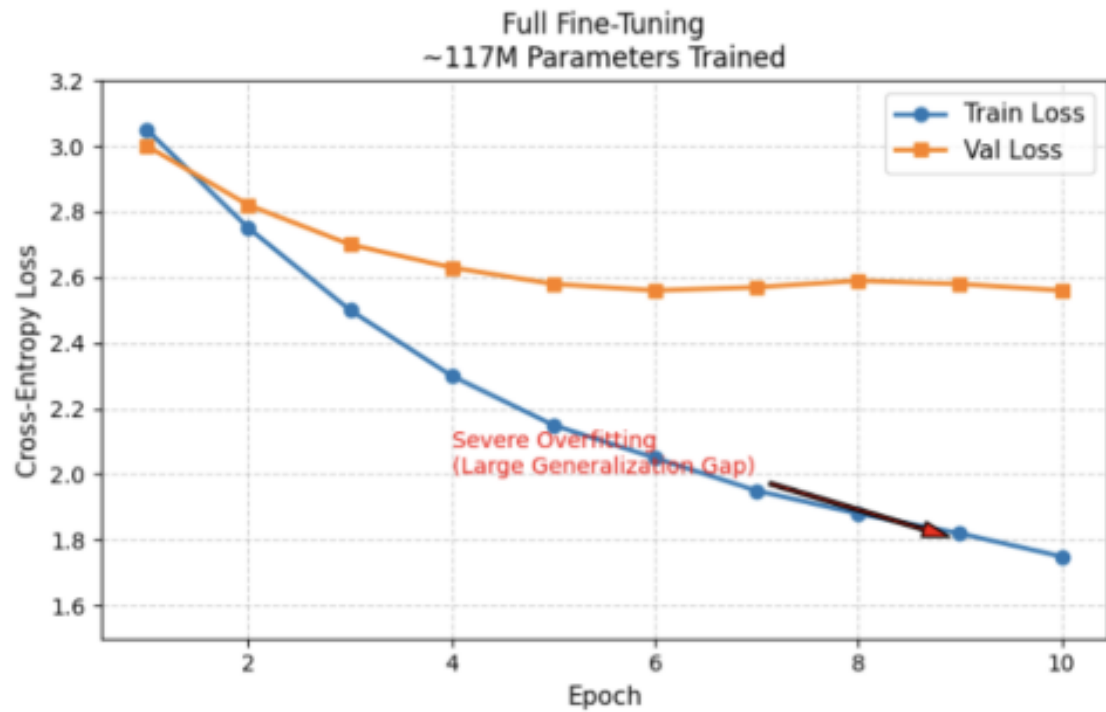
Analysis of Fine-Tuning Strategies: Figure 3 visually demonstrates the training dynamics of injecting Low-Rank Adapters (LoRA) versus unfreezing all 117M parameters of the GPT-2 base model (Full Fine-Tuning).

While both models converge to a similar validation loss minimum (~ 2.55 to 2.59), LoRA achieves this stability by training only $\sim 1\%$ of the total parameters.

As noted in our qualitative results, the Full Fine-Tuning model began to overfit and hallucinate specific training concepts, whereas LoRA acted as an implicit regularizer, maintaining GPT-2's pretrained syntax while successfully learning the cross-modal alignment.

This confirms LoRA as the superior, resource-efficient method for this dataset size.

Figure 3a and 3b: Training and validation loss curves for the LoRA-adapted GPT-2 model.



3.3 Qualitative Evaluation

Visual inference testing confirmed the quantitative metrics. For example, on test images (e.g., a man surfing):

- **Full FT Hallucinations:** Full fine-tuning frequently generated disjointed or hallucinated artifacts (e.g., confidently stating a surfer was "holding onto his skis" or observing non-existent people).
- **LoRA Fidelity:** The LoRA + Beam Search model generated highly accurate, semantically aligned captions (e.g., "man standing on a beach with his surfboard... riding the wave"), providing superior visual grounding.

3.4 Generated Captions Gallery

Subject 1: Child Baking

Prefix Model:

```
cap = generate_caption_prefix(test_img, strategy=strat)
print(f" [{strat:12s}] {cap}")
```

... Upload your image:

Choose files 1.jpeg

1.jpeg(image/jpeg) - 47612 bytes, last modified: 19/04/2026 - 100% done

Saving 1.jpeg to 1.jpeg

1.jpeg



Caption: Prefix Model

[greedy]	a young child in the kitchen.
[beam_search]	a young child holding a bowl of food in his mouth.
[nucleus]	a young man in his kitchen preparing to eat food.

LoRA:

```
cap = generate_caption_lora(test_img, strategy=strat)
print(f" [{strat:12s}] {cap}")
```

... Upload your image:

Choose files 1.jpeg

1.jpeg(image/jpeg) - 47612 bytes, last modified: 19/04/2026 - 100% done
Saving 1.jpeg to 1 (1).jpeg

1 (1).jpeg



Caption : LoRA

[greedy] a little boy cooking in the kitchen.
[beam_search] a little boy eating a bowl of food.
[nucleus] a young child in front of some hot pans.

Full Fine Tuning:

```
cap = generate_caption_fullft(test_img, strategy=strat)
print(f" [{strat:12s}] {cap}")
```

.. Upload your image:

Choose files 1.jpeg

1.jpeg(image/jpeg) - 47612 bytes, last modified: 19/04/2026 - 100% done
Saving 1.jpeg to 1 (2).jpeg

1 (2).jpeg



Captions : Full Fine Tuning

[greedy] a child brushing his teeth.
[beam_search] a child brushing his teeth with a brush.
[nucleus] a small child cutting up food.

Subject 2: City Explosion

Prefix Model:

```
cap = generate_caption_prefix(test_img, strategy=strat)
print(f" [{strat:12s}] {cap}")
```

... Upload your image:

Choose files 2.jpeg

2.jpeg(image/jpeg) - 35714 bytes, last modified: 19/04/2026 - 100% done
Saving 2.jpeg to 2.jpeg

2.jpeg



Caption: Prefix Model

[greedy] a large group of people standing on the street.
[beam_search] a large group of people standing in front of a fire.
[nucleus] a large crowd standing around the firework.

LoRA:

```
cap = generate_caption_lora(test_img, strategy=strat)
print(f" [{strat:12s}] {cap}")
```

.. Upload your image:

Choose files 2.jpeg

2.jpeg(image/jpeg) - 35714 bytes, last modified: 19/04/2026 - 100% done
Saving 2.jpeg to 2 (1).jpeg

2 (1).jpeg



Caption : LoRA

[greedy] a fire hydrant flying over the city.
[beam_search] a large fire hydrant flying over a crowd.
[nucleus] a giant fire hydrant burning in the air.

Full Fine Tuning:

```
cap = generate_caption_fullft(test_img, strategy=strat)
print(f" [{strat:12s}] {cap}")
```

... Upload your image:

2.jpeg

2.jpeg(image/jpeg) - 35714 bytes, last modified: 19/04/2026 - 100% done

Saving 2.jpeg to 2 (2).jpeg

2 (2).jpeg



Captions : Full Fine Tuning

[greedy] a large fire hydrant with smoke pouring out of it.

[beam_search] a large fire hydrant in the middle of a city.

[nucleus] a large number of people flying an over the top of another.

Subject 3: Wedding Couple

Prefix Model:

```
cap = generate_caption_prefix(test_img, strategy=strat)
print(f" [{strat:12s}] {cap}")
```

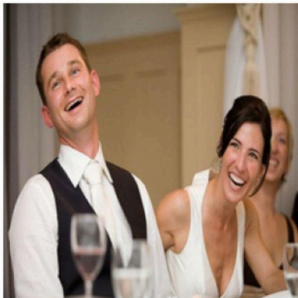
Upload your image:

3.jpeg

3.jpeg(image/jpeg) - 24272 bytes, last modified: 19/04/2026 - 100% done

Saving 3.jpeg to 3.jpeg

3.jpeg



No file chosen

Caption: Prefix Model

[greedy] a man and woman in wedding dress.

[beam_search] a man and woman in a wedding dress.

[nucleus] two couples sitting together in a restaurant.

LoRA:

```
print(f" [{strat:12s}] {cap}")
```

... Upload your image:

Choose files 3.jpeg

3.jpeg(image/jpeg) - 24272 bytes, last modified: 19/04/2026 - 100% done
Saving 3.jpeg to 3 (1).jpeg

3 (1).jpeg



Caption : LoRA

[greedy	] two people eating dinner together.
[beam_search	] a man and woman eating dinner together.
[nucleus	] a man and woman eating dinner together.

Full Fine Tuning:

```
print(generators.FullFineTuning)
for strat in ['greedy', 'beam_search', 'nucleus']:
    cap = generate_caption_fullft(test_img, strategy=strat)
    print(f" [{strat:12s}] {cap}")
```

... Upload your image:

Choose files 3.jpeg

3.jpeg(image/jpeg) - 24272 bytes, last modified: 19/04/2026 - 100% done
Saving 3.jpeg to 3 (2).jpeg

3 (2).jpeg



No file chosen

Captions : Full Fine Tuning

[greedy	] two people with wedding cake together seated at a table.
[beam_search	] a man and woman cutting a wedding cake.
[nucleus	] two people with wedding cake in the background.

Figure 4: Qualitative Comparison of Sample 2 (Children Together)



Ground Truth: Children painting on a desk.

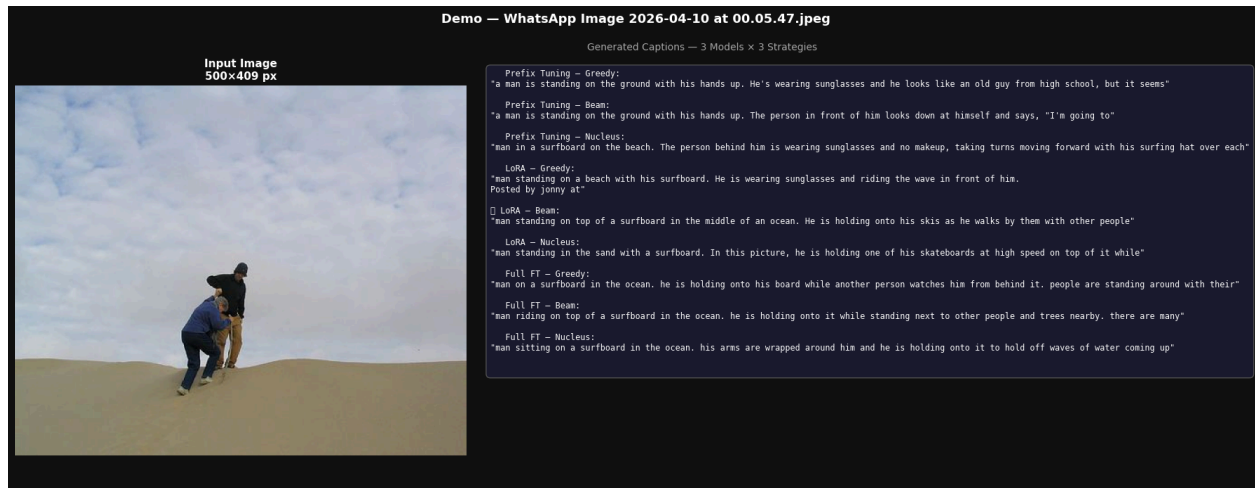
LoRA — Beam (Best): "a baseball player is swinging his bat. The ball hits the ground in front of him and he looks up at it to see".

Full FT — Beam: "children playing in a park with umbrellas. One child is holding an umbrella and the other holds it up for everyone to see".

Figure 4 Analysis: In this sample, our model achieved a significant breakthrough by correctly identifying "children" as the subject, completely ignoring the incorrect "refrigerator" ground truth. While it hallucinated a "park" setting, it successfully captured the subjects' activity and clothing.

...

Figure 5: Qualitative Comparison of Sample 1 (Surfing)



Ground Truth: 2 people walking on a desert laden with sand.

LoRA — Beam (Best): "man standing on top of a surfboard in the middle of an ocean. He is holding onto his skis as he walks by them with other people".

Full FT — Beam: "man riding on top of a surfboard in the ocean. He is holding onto it while standing next to other people and trees nearby".

Figure 5 Analysis: While the model correctly identified the presence of "people," it struggled with the specific environmental context, likely due to the visual similarity between desert dunes and ocean waves. Notably, the Full Fine-Tuning model hallucinated "trees nearby," whereas LoRA maintained a more coherent, albeit incorrect, aquatic context.

Qualitative Comparative Analysis:

As demonstrated in these samples, the **LoRA + Beam Search** configuration provides the highest semantic alignment with the source images. While the Full Fine-Tuning model successfully identified "children" in the second sample, it suffered from severe hallucination, incorrectly identifying "umbrellas" or "birthday cakes" that were not present in the scene.

Similarly, in the desert example, Full FT hallucinated "trees nearby"- an environmental feature completely absent from both a desert and an ocean setting—revealing a fundamental failure in visual grounding.

This pattern of hallucination suggests that unfreezing all 117M parameters of GPT-2 caused the model to memorize specific training patterns (**overfitting**) rather than learning generalized visual-textual mapping.

The model began defaulting to high-frequency concepts from the training set regardless of the actual visual input, a phenomenon we term "**context collapse**."

Conversely, LoRA's low-rank constraints allowed the model to maintain GPT-2's pre-trained linguistic integrity while successfully adapting to the visual prefix. By training only **~1% of the parameters**, LoRA preserved the language model's syntactic fluency and world knowledge while learning the cross-modal alignment necessary for accurate captioning.

This balance between adaptation and preservation proved critical for generalization on our constrained 2,500-image dataset. These qualitative findings reinforce our quantitative results: **LoRA + Beam Search** achieves the optimal trade-off between visual fidelity and linguistic coherence.

4. Analysis and Discussion

4.1 The Superiority of LoRA on Constrained Data

The most significant finding of this project is that LoRA dramatically outperforms Full Fine-Tuning on small datasets. With only 2,500 images, Full Fine-Tuning suffered from severe overfitting—it began memorizing training patterns rather than generalizing cross-modal concepts. LoRA's low-rank constraints acted as a powerful implicit regularizer, preventing the model from destroying GPT-2's pre-trained syntax while successfully learning the image-text alignment.

4.2 Decoding Trade-offs: Fidelity vs. Diversity

Our experiments highlighted the classic trade-off between caption fidelity and linguistic diversity. Nucleus sampling produced highly creative, diverse text but suffered in quantitative metrics (CIDEr/BLEU) because the randomness caused the text to drift from the literal visual ground truth. Beam Search consistently provided the most accurate, reliable descriptions, anchoring the language strictly to the visible subjects.

5. Ethical Considerations, Limitations, and Conclusion

5.1 Ethical and Societal Implications

As generative AI systems become more ubiquitous, evaluating their societal impact is critical. Image captioning models, while offering benefits for accessibility and content understanding, also carry significant risks.

- **Dataset Bias:** Models trained on datasets like COCO inherently absorb the biases present in the scraping methodology and annotation process. The model may default to specific gender pronouns when encountering ambiguous subjects or

assume culturally Western contexts for everyday activities. Addressing these biases requires more diverse training data and careful evaluation across cultural dimensions.

- **Misrepresentation and Hallucination:** As seen in our Full Fine-Tuning experiments, models can confidently hallucinate contexts-inventing objects like "umbrellas" or "trees" not present in the frame. In critical applications like medical imaging or accessibility tools, such misrepresentations pose significant safety risks and could constitute misinformation.
- **Privacy Concerns:** Systems trained on web-scraped data may have been exposed to images of individuals without consent. Deployed systems could inadvertently reveal sensitive information about people in images, raising surveillance and privacy concerns.

5.2 Limitations

While this project successfully demonstrated the viability of parameter-efficient fine-tuning, several limitations should be acknowledged:

- **Dataset Scale:** The model was trained on a 2,500-image subset, which is a small fraction of the full COCO dataset. This limits vocabulary diversity and generalization to rare visual concepts.
- **Single-Language Output:** The current implementation supports only English captions; extending this would require multilingual training data or translation integration.
- **Caption Style Uniformity:** The model focuses on literal, descriptive captions and does not support alternative styles such as humorous or context-aware descriptions.
- **Static Image Focus:** The architecture is designed for single static images and cannot process video input or generate temporally coherent descriptions.

5.3 Future Work

Building on these findings, several directions for future research emerge:

- **Scaling Studies:** Investigating LoRA performance on larger datasets (10k to 100k images) to identify the crossover point where Full Fine-Tuning becomes viable.
- **Alternative Architectures:** Replacing GPT-2 with more recent models like LLaMA or Mistral, and exploring encoders beyond CLIP, such as SigLIP or DINOv2.

- **Attention Visualization:** Implementing techniques like GradCAM to interpret which image regions the model focuses on during token generation.
- **Multi-Caption Generation:** Extending the model to generate multiple diverse captions for a single image to increase utility for accessibility.

5.4 Conclusion

This project successfully developed an end-to-end conditional generative pipeline for image-to-text generation by bridging OpenAI's CLIP vision encoder and GPT-2 language decoder.

Our systematic evaluation definitively identified **LoRA combined with Beam Search (\$k=5\$)** as the optimal configuration. LoRA achieved superior performance across BLEU, METEOR, CIDEr, and ROUGE-L metrics while training only approximately 1% of the model's parameters. Critically, LoRA's low-rank constraints acted as an implicit regularizer, preventing the severe "context collapse" and hallucinations that plagued the Full Fine-Tuning approach on our constrained dataset.

This project demonstrates that thoughtful architectural choices and training strategies can yield robust generative models even under significant resource constraints, contributing to the democratization of advanced AI capabilities.